

Skill Week :-3

Task :-1

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        // Display prompt
        printf("OSSP> ");
        fflush(stdout);

        // Read user input
        if (fgets(command, sizeof(command), stdin) == NULL)
        {
            printf("\n");
            break;
        }

        // Remove newline character
        command[strcspn(command, "\n")] = '\0';

        // Handle empty input
        if (strlen(command) == 0)
        {
            continue;
        }
    }
}
```

```

// Handle exit condition
if (strcmp(command, "exit") == 0)
{
    printf("Exiting shell...\n");
    break;
}

// Display entered command
printf("You entered: %s\n", command);
}

return 0;
}

```

Output:-

```

anuan@Anushka:~$ nano main_loop.c
anuan@Anushka:~$ gcc main_loop.c -o main_loop
anuan@Anushka:~$ ./main_loop
OSSP> hello
You entered: hello
OSSP> ls
You entered: ls
OSSP> pwd
You entered: pwd
OSSP> test
You entered: test
OSSP> exit
Exiting shell...

```

Task:-2

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

#include <stdlib.h>

```

```
int main()
{
    int fd1, fd2;
    char buffer[100];
    ssize_t n;

    // Open File1.txt for reading
    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 < 0)
    {
        perror("Error opening File1.txt");
        return 1;
    }

    // Open File2.txt for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 < 0)
    {
        perror("Error opening File2.txt");
        close(fd1);
        return 1;
    }

    // Read from File1.txt and write to File2.txt
    while ((n = read(fd1, buffer, sizeof(buffer))) > 0)
    {
        if (write(fd2, buffer, n) != n)
        {
            perror("Error writing to File2.txt");
        }
    }
}
```

```
        close(fd1);
        close(fd2);
        return 1;
    }
}

if (n < 0)
{
    perror("Error reading File1.txt");
}
else
{
    printf("File copied successfully!\n");
}

// Close both files
close(fd1);
close(fd2);

return 0;
}
```

Output:-

```

anuan@Anushka:~$ nano File1.txt
anuan@Anushka:~$ nano file_copy.c
anuan@Anushka:~$ gcc file_copy.c -o file_copy
anuan@Anushka:~$ ./file_copy
File copied successfully!
anuan@Anushka:~$ ls
File1.txt  file_copy.c      hello          process.c
File2.txt  file_copy.c.save hello.c         process.c.save
OSSP      fork_exec        main_loop      program.c
Practical  fork_exec.c      main_loop.c    pwd
Skill      fork_exec.c.save process
file_copy  fork_exec.r      process.
anuan@Anushka:~$ cat Filed.txt
cat: Filed.txt: No such file or directory
anuan@Anushka:~$ cat File1.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.
anuan@Anushka:~$ cat File2.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.
anuan@Anushka:~$ |

```

Task :-3

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX 100
```

```
int main()
```

```
{
```

```
    char buffer[MAX];
```

```
    int index = 0;
```

```
    char ch;
```

```
    printf("Simple Keyboard Input Program\n");
```

```
    printf("Type a command (type 'exit' to quit)\n");
```

```
    while (1)
```

```
    {
```

```
        printf("OSSP> ");
```

```
        index = 0;
```

```
        while (1)
```

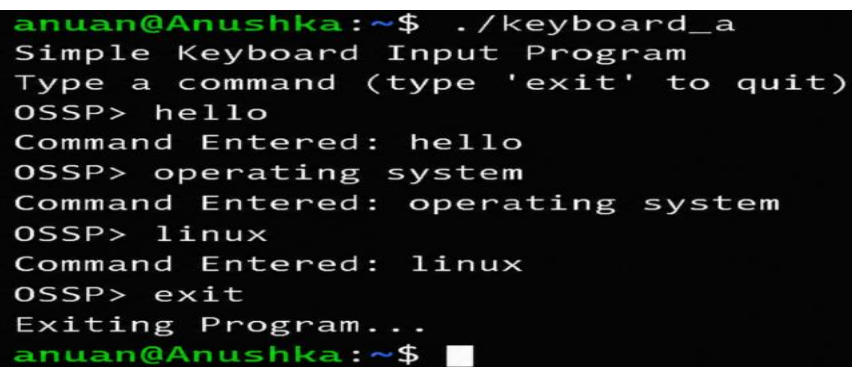
```
{  
    ch = getchar();  
  
    // Enter key  
    if (ch == '\n')  
    {  
        buffer[index] = '\0';  
        break;  
    }  
  
    // Backspace key  
    else if (ch == 127 || ch == '\b')  
    {  
        if (index > 0)  
        {  
            index--;  
            printf("\b \b");  
        }  
    }  
  
    // Store normal characters  
    else  
    {  
        if (index < MAX - 1)  
        {  
            buffer[index++] = ch;  
        }  
    }  
}  
  
if (strcmp(buffer, "exit") == 0)  
{  
    printf("Exiting Program...\n");
```

```
        break;
    }

    printf("Command Entered: %s\n", buffer);
}

return 0;
}
```

Output:-



```
anuan@Anushka:~$ ./keyboard_a
Simple Keyboard Input Program
Type a command (type 'exit' to quit)
OSSP> hello
Command Entered: hello
OSSP> operating system
Command Entered: operating system
OSSP> linux
Command Entered: linux
OSSP> exit
Exiting Program...
anuan@Anushka:~$
```